Engagement Pulse | Azure deployment & Entra ID

Proposed approach · Current application runs locally with Docker Compose and JWT authentication.

1. Deployment approach

I would keep the existing containers and move each responsibility to an Azure managed service. The goal is to deploy the application without changing its core architecture.

Current component	Proposed Azure service
Next.js frontend	Azure Container Apps
Django API + WebSockets	Azure Container Apps (Daphne, HTTPS/WSS)
Dramatiq worker	Separate Azure Container App
PostgreSQL	Azure Database for PostgreSQL
Redis (jobs + live updates)	Azure Managed Redis
CSV file storage	Azure Blob Storage (instead of Docker volume)

2. Moving sign-in to Microsoft Entra ID

I would replace the current email/password login with Microsoft Entra ID using OpenID Connect. The frontend redirects users to Microsoft for sign-in, and the backend validates the tokens it receives.

What stays the same: Django continues to enforce Consultant / Client Viewer permissions and organisation access from PostgreSQL. Entra ID confirms who the user is; it does not decide which client data they can see.

Migration: Register the application in Entra ID, configure sign-in and token validation, then link Entra identities to existing application users using stable identity IDs rather than email alone.

3. What I would validate before production

Verify sign-in and token expiry, tenant isolation, WebSocket access, worker connectivity and CSV uploads. Add managed secrets, HTTPS, monitoring, backups and health checks. Scale API instances and workers independently as traffic grows.

Scope: *This is a proposed Azure design, not a deployed environment. My hands-on cloud experience is primarily AWS; the design maps the existing application to equivalent Azure services.*